Nanoscale effects leading to non-Einstein-like decrease in viscosity

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Nanoparticles have been shown to influence mechanical properties; however, transport properties such as viscosity have not been adequately studied. This might be due to the common observation that particle addition to liquids produces an increase in viscosity, even in polymeric liquids, as predicted by Einstein nearly a century ago. But confinement and surface effects provided by nanoparticles have been shown to produce conformational changes to polymer molecules, so it is expected that nanoparticles will affect the macroscopic viscosity. To minimize extraneous enthalpic or other effects, we blended organic nanoparticles, synthesized by intramolecular crosslinking of single polystyrene chains, with linear polystyrene macromolecules. Remarkably, the blend viscosity was found to decrease and scale with the change in free volume introduced by the nanoparticles and not with the decrease in entanglement. Indeed, the entanglements did not seem to be affected at all, suggesting unusual polymer dynamics.

The rheology of particulate suspensions has a long history beginning with Einstein[1], who demonstrated that the viscosity increase that brownian particles impart is solely a function of the particle volume fraction (ϕ) and the viscosity of the suspending liquid. This prediction has been experimentally confirmed many times[2,3], when a simple or monomeric suspending fluid is used. Suspension of particles in polymeric liquids produces a similar increase in viscosity[4] even when clay–polymer nanocomposites are studied[5]. In the clay system, the particles are nanoscopic in one direction and significantly larger than the polymeric liquids' structural length scale in the other two.

Siegel[6] described how mesoscopic physics or nanotechnology can change the macroscopic physical properties of materials. Indeed, Mayo *et al.*[7,8] demonstrated how ceramic materials can change from brittle to ductile through grain-boundary sliding when cluster sizes of less than 15 nm are present. In a similar vein, Roberts *et al.*[9] blended small silicate clusters (with a radius, a, of about 0.35 nm) with linear polymers and observed a decrease in viscosity, again pointing to the unusual effect that nanoscale processes and materials might provide. However, these particles are almost molecular in size and approach the monomer length scale, so interpretation may be clouded by differentiating between a plasticizer molecule and a nanoparticle. Roberts *et al.* state that these small clusters behave more like a solvent than a particle, akin to the function of a plasticizer, so it is unclear whether nanoscopic effects can produce unexpected flow properties.

To investigate this issue unequivocally, we suspended polystyrene (PS) nanoparticles in linear PS to eliminate enthalpic interactions. Further, because the particles and polymer are essentially the same material and have similar refractive indexes, dispersion forces are minimized[3,10] creating a hard-sphere system in which only excluded volume forces are apparent. This is therefore an ideal system in which to study and delineate nanoscopic effects without the nuisance of enthalpic or other forces.

In this article, spherical nanoparticles with low polydispersity ($a \approx 3$–5 nm), synthesized by a method involving intramolecular collapse[11], are blended with linear PS macromolecules of approximately the same molecular dimensions. For example, at a relative molecular

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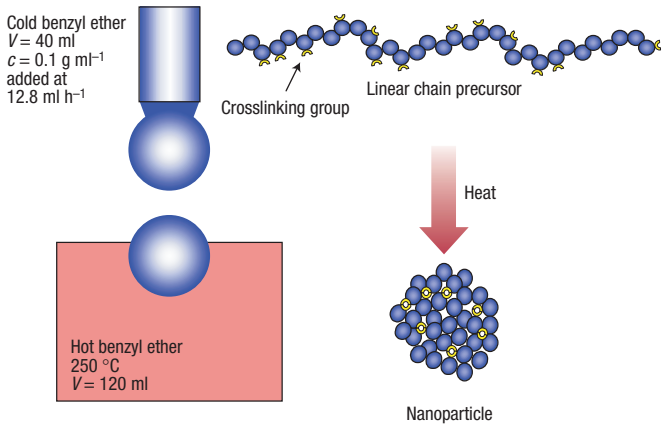

Figure 1 Diagram of the intramolecularly crosslinked nanoparticles produced from linear-chain precursors having pendent crosslinking groups. Varying the amount of crosslinker permits the fabrication of lightly crosslinked (2.5 mol% crosslinker) or tightly crosslinked (20 mol% crosslinker) nanoparticles. Nanoparticles were produced by dripping a solution into hot solvent to activate the crosslinking process. Typical conditions of volume (V), concentration (c), flow rate and temperature are given.

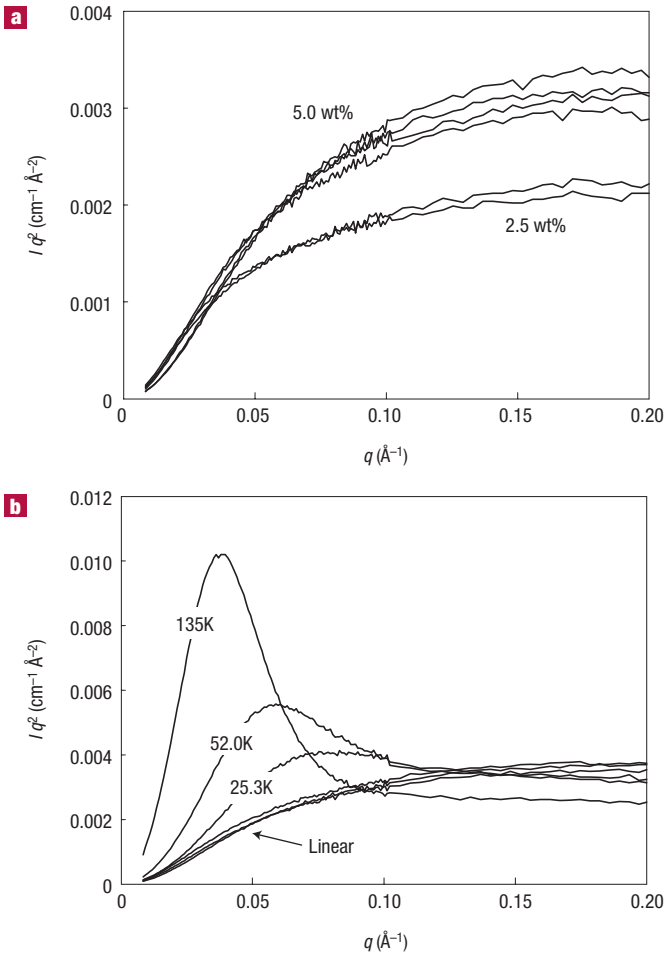

Figure 2 SANS spectrum for lightly and tightly crosslinked nanoparticles, demonstrating the particle-like nature of the nanoparticles when sufficient crosslinking is present. a, Kratky plot of SANS intensity (I) multiplied by the square of the wavevector (q) as a function of q for solutions of linear precursor polymer or lightly crosslinked nanoparticle containing 2.5 mol% crosslinker. The lower concentration data of 2.5 wt% are for the M_r 158K linear-chain precursor polymer and nanoparticle, whereas the higher concentration curves, 5.0 wt%, are for the precursor polymer and nanoparticle with nominal M_r values of 24.5K and 158K. These data show that the scattering from the lightly crosslinked nanoparticles is equivalent to that from a linear polymer. All data were measured in deuterated THF solvent at 35 °C. The plateau near 0.2 Å^{-1} does not scale linearly with concentration because of concentration effects. **b**, Kratky plot for the tightly crosslinked nanoparticles containing 20 mol% crosslinker of various sizes (M_r 25.3K, 52K and 135K) compared with the linear-chain precursors (curves labelled Linear). These nanoparticles demonstrate considerably different scattering behaviour from that of the linear precursors, with the curve maxima revealing particle-like behaviour. The same solvent and temperature as in **a** were used; the concentration was 5 wt%. The maximum position does not depend significantly on the nanoparticle concentration.

mass of 52,000 (M_r 52K), with every fifth monomer unit crosslinked on average, the nanoparticles are of order 3–5 nm in radius and comparable to the linear polymer radius of gyration (R_g), 7.5–15 nm. Two types of nanoparticle were used, lightly crosslinked (2.5 mol% crosslinker) and tightly crosslinked (20 mol% crosslinker; see Fig. 1), which were synthesized with a variety of molecular masses and hence molecular sizes. The procedure is relatively simple and is robust enough to produce gram quantities of nanoparticles with controlled size and crosslinking densities from linear precursor chains[11].

Crosslinking imposes a particle-like nature to the linear precursor chains that small-angle neutron scattering (SANS) can measure, as shown in Fig. 2. Lightly crosslinked nanoparticles exhibit very similar scattering behaviour to the linear precursor chains, whereas the tightly crosslinked nanoparticles demonstrate a peak in the Kratky plot, which is indicative of particle-like behaviour. The similarity of the lightly crosslinked nanoparticles to the linear polymer, at least in their segment density profiles, translates to a likeness in their rheological properties. This does not invalidate the hypothesis of Antonietti *et al.*[12] that crosslinked microgels and linear polymers are rheologically alike; indeed, it may strengthen it because even light crosslinking will produce cycles within the molecule.

A high wavevector plateau is present for the tightly crosslinked nanoparticles after a local maximum, unlike the observed behaviour for a homogeneous density hard sphere[13]. This phenomenon was seen by Tande *et al.*[14] for poly(benzyl ether) dendrimers, which is suggestive of both particle and gaussian chain behaviour. The ratio, P, of the radius of gyration to the viscosimetric radius (determined from the intrinsic viscosity) is an important quantity in measuring segment density profiles and physics[15]. The dendrimer system displayed P values that decreased from the value for a gaussian coil (1.2–1.8, depending on solvent conditions) to that for a homogenous hard sphere ($0.775 = \sqrt{(3/5)}$) with increasing generation number (molecular mass). The tightly crosslinked nanoparticles show a similar phenomenon in a good solvent (tetrahydrofuran (THF) at 35 °C), with P values of 1.24 (M_r 25.3K), 1.15 (M_r 52.0K) and 0.93 (M_r 135K) that approach the hard-sphere value with increasing molecular mass. However, the constant-segment-density profile is not achieved, which contributes to the plateau in the Kratky plot. Thus, the tightly crosslinked nanoparticles are intermediate between a polymer chain and a sphere of homogeneous density, and remarkably they resemble the gaussian soft sphere[16]. Indeed, the scattering form factor for the gaussian soft sphere looks very similar to that for the tightly crosslinked nanoparticles.

A critical feature of polymer–particle systems is the possible poor dispersion resulting from nanoparticle agglomeration and/or

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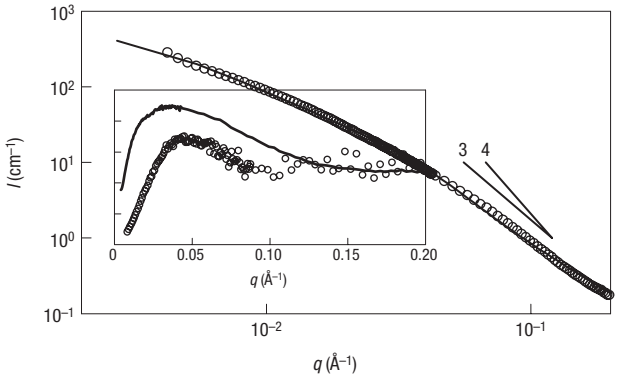

Figure 3 SANS spectrum for tightly crosslinked nanoparticles (M_r 52K) blended with linear deuterated polystyrene (M_r 63K) at 50 wt% with the incoherent background subtracted from the intensity. The solid line represents a hard-sphere fit, including polydispersity, and a lorentzian ($I(0)/[1 + \xi^2 q^2]$, $\xi = 29.5 \pm 0.2$ nm) and $I(0) = 623 \pm 7$ cm^{-1}) to account for the deuterated polystyrene chains' conformation. The mean nanoparticle phase's radius from the hard-sphere fit is 2.66 ± 0.01 nm, the polydispersity (standard deviation divided by mean) is 0.942 ± 0.001 and the volume fraction is 0.45. Power laws of 3 and 4 are shown; neither is satisfactory, suggesting that a fractal network is not present. The inset is a Kratky plot of the same data (solid line) compared with data for the same nanoparticles in dilute solution (5.0 wt% in THF at 35 °C) demonstrating increased scatterer size. The intensity is in arbitrary units and the curves were shifted to overlap at high q.

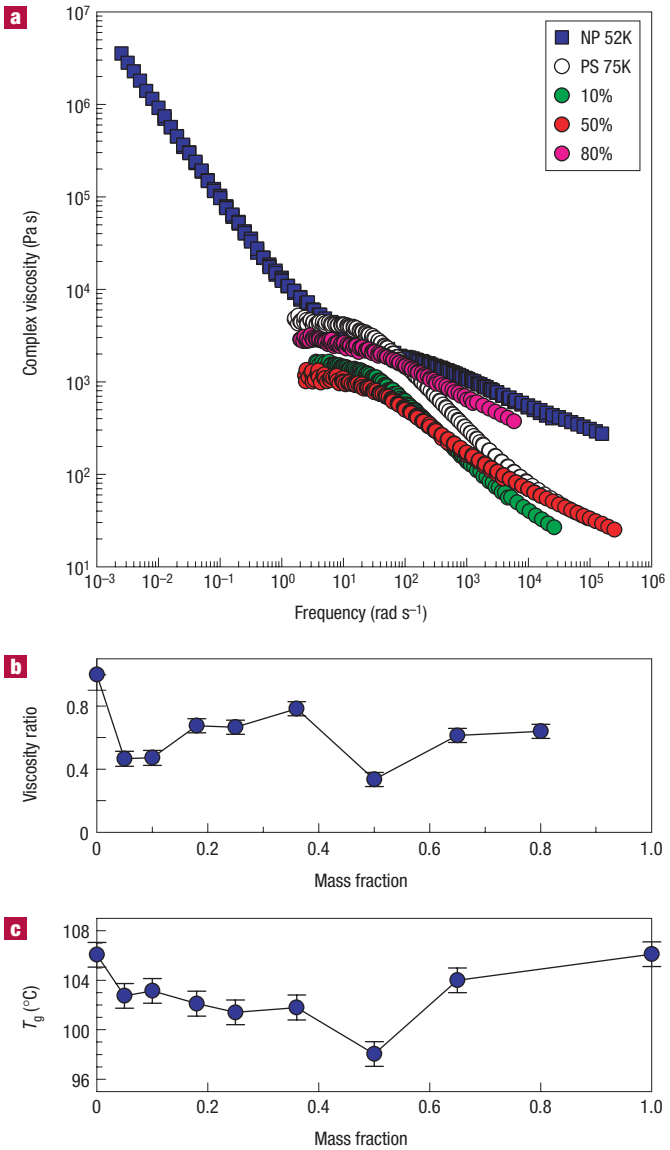

Figure 4 Viscosity and glass-transition temperature (T_g) for M_r 52K tightly crosslinked nanoparticles (NP 52K), M_r 75K linear polystyrene (PS 75K) and their blends. a, Plot of viscosity against frequency (measured with a Rheometrics ARES rheometer) for the undiluted nanoparticles (NP 52k), undiluted linear PS (PS 75k) and selected blends containing 10 wt%, 50 wt% and 80 wt% nanoparticles. Note that the pure nanoparticle system shows no terminal viscosity. Various temperatures were used and time–temperature superposition was followed; the reference temperature was 170 °C. **b**, Plot of terminal viscosity of the blends divided by the viscosity of undiluted M_r 75K linear PS against the mass fraction of M_r 52K tightly crosslinked nanoparticles, at 170 °C. **c**, T_g measured by differential scanning calorimetry (TA Instruments Q1000) for the blends, showing a general decrease with nanoparticle addition up to moderate volume fractions. Note that the T_g values for the two pure components are equal.

interpenetration in the tightly crosslinked nanoparticle–linear PS blends. Indeed, agglomeration might be expected owing to depletion flocculation; however, the results of Cosgrove *et al.*[17] clearly show that silica nanoparticles 16 nm in diameter are stable in poly(ethylene oxide)/water solution up to high concentrations of polymer and particles. In this case, the particle half-separation distance is much lower than the polymer R_g, (which is 10–35 nm) in many of their experiments, and no evidence of agglomeration is seen even up to particle concentrations of 18 vol.%, which must be due to the nanoparticle size and scale.

The results in Fig. 3 for a 50 wt% blend show that the blends used in our work are well dispersed, because the intensity profile shows no upturn at small wavevector and no fractal-like regime at high wavevector; compact objects are therefore present. A gaussian plateau occurs and a fit for a hard-sphere model together with a lorentzian, to account for the linear polymer conformation, yields an average particle size of 2.7 nm. From the molecular mass of the nanoparticles and this radius one can calculate that the nanoparticles have collapsed to a density equivalent to bulk PS. Although the model results are not conclusive, it is clear that the M_r 52K tightly crosslinked nanoparticles have collapsed to a smaller size than in dilute solution, in which the hydrodynamic radius is 4.8 nm in THF. Clearly, the SANS results and modelling are suggestive of nanoparticles collapsed to a density equivalent to bulk PS, yet a fractal-like nanoparticle network is not present. Further, a highly contorted linear chain is suggested by the lorentzian mesh length of about 30 nm. We recognize that a mesh length of 30 nm is extreme, and this is justified below.

The (complex) viscosity was measured as a function of frequency, as shown in Fig. 4a. Different temperatures were used, and time–temperature superposition was followed to create a master curve at a reference temperature of 170 °C. The undiluted tightly crosslinked nanoparticles do not exhibit a terminal viscosity, even at very low frequencies. It is clear that all blends, even up to 80 wt%, show a terminal (zero frequency) viscosity.

The terminal viscosity was determined from the data and was found to decrease when nanoparticles were added (Fig. 4b), even though the pure M_r 52K nanoparticles show no terminal viscosity[12]. A concentration of 50 wt% demonstrates a fourfold decrease in viscosity, whereas predictions from either the Einstein or Kreiger–Dougherty[18] models suggest a large increase in viscosity. For example, use of the

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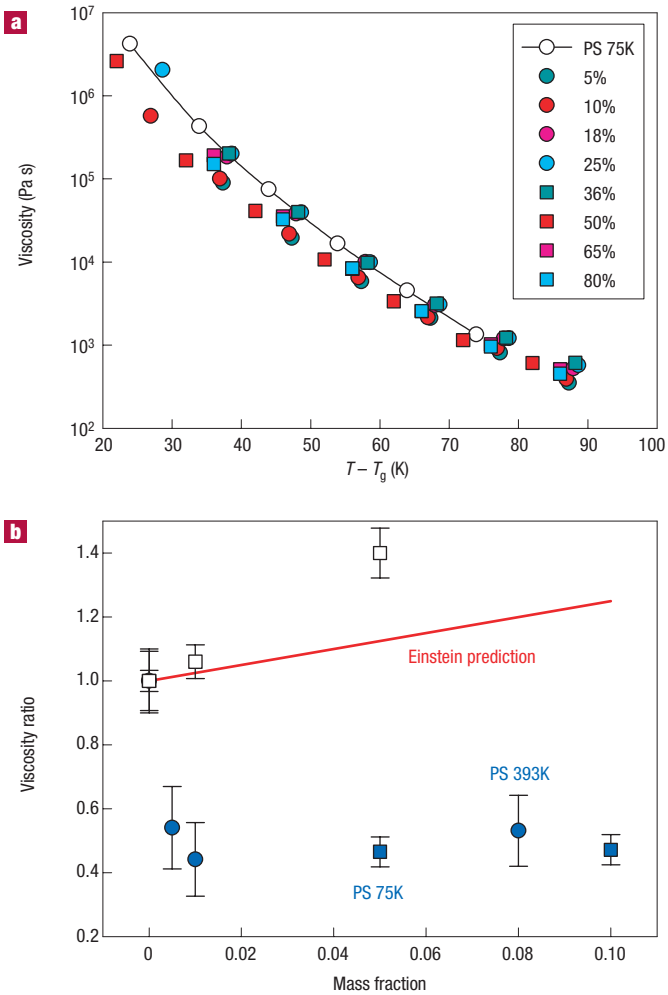

Figure 5 Change in viscosity with temperature and nanoparticle concentration. **a**, Viscosity of M_r 75K undiluted polystyrene (PS) and nanoparticle blends as a function of $T - T_g$. Open circles show the viscosity of the undiluted PS; the other symbols represent data for tightly crosslinked M_r 52K nanoparticles dispersed in the same PS at various mass concentrations (5–80%). The measurement temperature range for all systems was 140–190 °C, with some extending down to 130 °C. There is no clear trend with concentration and it is proposed that the viscosity is a unique function of distance from T_g. **b**, Comparison of the Einstein prediction with results for the lower-concentration blends containing the tightly crosslinked M_r 52K nanoparticles in M_r 75K linear PS (blue squares) and M_r 393K linear PS (blue circles) at 170 °C. It is clear that a hydrodynamic contribution to the viscosity, in the contemporary sense, is not provided by the nanoparticles. On the contrary, the addition of 1.6-μm diameter crosslinked PS microspheres shows the conventional viscosity increase with particle addition to M_r 75K linear polystyrene (white squares).

Kreiger–Dougherty model estimates that the ratio of the viscosity of the suspension to that of the pure polymer should be about 10 rather than the measured 0.2–0.3. Further, the 80 wt% nanoparticle blend displays a terminal viscosity when a gel-like response is expected at such high concentration[3]. This result points to change in shape of the nanoparticles, and possible agglomeration and polymer interpenetration. However, if significant agglomeration were present there should be an increase in viscosity.

Slip or inhomogeneous flow could account for the decrease in viscosity[19]. This was checked by using different gaps between the parallel plates used to characterize the viscosity: gaps of 100–500 μm were used with the 10 wt% blend and no gap effect was seen, so slip and inhomogeneous flow are not a factor in the rheological measurements.

Supplementary testing with the blends shows a concomitant decrease in glass-transition temperature (T_g) as nanoparticles are added, despite each component having almost identical T_g values (see Fig. 4c). It should be noted that T_g does not change when the lightly crosslinked nanoparticles are added to the linear PS: both have equivalent T_g values, and the viscosity is intermediate between the two components following a normal mixing rule based on the weight-average molecular mass. Starr *et al.*[20] show a decrease in polymer T_g near nanoparticles when excluded-volume interactions are present; however, if an attractive potential is present T_g is predicted to increase. As discussed above, we have an ideal system that can interact only through excluded volume. Yet we believe that our system is different from the simulations performed by Starr *et al.* because they did not specifically simulate confinement effects. However, similar results of a change in T_g are seen in confined thin polymer films[21], although we admit here that this is a complicated and rapidly advancing area. We propose that nanoparticle confinement effects produce the decrease in T_g because the interparticle half-gap, h, achieves quite small values. Further, all the data in the present study are at concentrations where h is less than or equal to R_g.

The use of a simple relation ($h/a \approx [\phi_m/\phi]^{1/3} - 1$, $\phi_m = 0.638$; ϕ_m is the maximum random packing fraction) or an empirical relation developed from the data of Cosgrove *et al.*[17] ($h/a \approx [3.11/\phi]^{1/4.34} - 1$, $0.18 > \phi > 0.03$) yields $h = 0.2$–1.4 nm for a volume fraction of 0.5. A half-gap of order 1 nm is much smaller than the R_g for the linear polymer and approaches the monomer length scale, thus producing severe chain confinement and distortion. Note that the above relations do not include the volume of the interstices between the particles, which is substantial, thereby allowing some volume for the polymer to occupy. However, the polymer molecule is distorted, as confirmed through the SANS modelling in Fig. 3, in which the lorentzian mesh length is of order 30 nm. This is an exceptional value and can be rationalized by realizing that the nanoparticle volume fraction is 0.5, making the volume ratio of nanoparticle (V_{NP}) to linear polymer (V_{PS}) equal to unity. The volume ratio is equal to the ratio of length scales to the third power ($[d_{NP}/d_{PS}]^3$); the mesh length, ξ, is $d_{NP} + d_{PS}$. Realizing that the polymer length scale is of the order of twice its radius of gyration (about 7 nm), one finds $\xi = 28$ nm, in agreement with the model fitting parameter. This clearly indicates some degree of distortion.

Both the virgin linear polymer and its blend with tightly crosslinked nanoparticles follow time–temperature superposition[22] with similar shift factors. The terminal viscosity of the pure linear polymer is plotted in Fig. 5a as a function of temperature above T_g; the usual decrease in viscosity with increasing temperature is apparent. Graphing the blends' viscosity in a similar manner, thereby accounting for the decrease in T_g, shows that the viscosity is a unique function of temperature above T_g.

Nanoparticles apparently do not contribute to the hydrodynamics of highly entangled linear PS molecules with M_r values of 75K and 393K, as shown in Fig. 5b. If they did, the viscosity would follow Einstein's model[1] at low concentration or Batchelor's correction for moderate concentration[23], and it is shown graphically that these relations are not followed. Larger particle systems in polymer melts clearly produce an increase in viscosity[4], so our results are a unique product of a nanosystem. To expand on this point we measured the viscosity of latex microspheres (PS crosslinked with divinylbenzene) 1.6 μm in diameter blended with M_r 75K linear PS. The results are shown in Fig. 5b: a clear viscosity increase is seen, in agreement with previous results.

Thus, we expect that the nanoparticles affect a physical process within the polymer melt. On the basis of the increased free volume that silica nanoparticles produce in poly(4-methyl-2-pentyne)[24] and that expanded PS coils have a 45 °C decrease[25] in T_g, as well as our results in Fig. 5a, we investigate whether nanoparticles can induce an increase in free volume. It is simple to determine that if a spherical

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excluded volume of thickness Δ exists around each nanoparticle, the fractional free volume is increased by approximately $3\phi\Delta/a$ assuming that the continuous phase's (that is, linear PS's) free volume does not change in the bulk. The fractional free volume increases by order 0.01, a significant change[22], for a 0.1 nm shell thickness and the volume fractions used in this study. This increase in free volume could explain the decrease in T_g and apparently the decrease in viscosity as well.

On the basis of the work of Merkel *et al.*[24] it might be expected that free volume is a critical factor in determining the rheological behaviour; however, the concepts of entanglements[22] or reptation[26] have not been addressed. The terminal (zero shear) viscosity for all nanoparticle concentrations is lower than that of undiluted polymer, as shown in Figs 4 and 5. However, concentrations above about 18 wt% have a viscosity at large frequencies higher than or approaching that of the virgin polymer, and the 5 wt% and 10 wt% blends have a viscosity that is always lower than the pure polymer. The 80 wt% blend exhibits minimal shear thinning and a large viscosity at high frequencies (see Fig. 4a), yet the loss modulus is always larger than the storage, which is indicative of enhanced dissipation. This is in contrast with the undiluted nanoparticle system that has an equilibrium storage modulus at small frequencies. These are unusual effects and could indicate constrained polymer chains as well as entanglement or reptative changes.

However, drawing from the scaling theory presented by deGennes[26] and furthered by Doi and Edwards[27], one finds that the terminal viscosity should scale as $G_N^0 t_R(M/M_e)$, where G_N^0 is the plateau modulus, t_R is the Rouse time scale, M is the polymer molecular mass and M_e is the mass between entanglements. Further, simple rubber elasticity[22] shows that the plateau modulus is given by a simple combination of the density (ρ), gas constant (R) and temperature (T) as well as the entanglement molecular mass, $\rho RT/M_e$. If the viscosity reduction were caused by an increase in M_e, representing a decreasing number of entanglements, one would expect both the plateau modulus and the terminal viscosity to decrease. This does not seem to be so. No clear trend of plateau modulus with nanoparticle concentration is seen, and its average for all concentrations is approximately the same as that for the pure polymer. For example, the plateau modulus for M_r 393K linear PS is $(1.7 \pm 0.1) \times 10^5$ Pa at 170 °C, and the average values for the polymer–nanoparticle composites are $(1.5 \pm 0.3) \times 10^5$ Pa and $(1.7 \pm 0.2) \times 10^5$ Pa for the M_r 52K and 135K tightly crosslinked nanoparticle blends, respectively. Further, at constant concentration (5–10 wt%) of the M_r 52K tightly crosslinked nanoparticles, the viscosity scales with the linear polymer molecular mass to the 3.5 power, whereas undiluted linear polymer shows a scaling of 3.6 for the molecular masses studied. These are power laws close to that found in the literature for entangled linear polymers[22]; if the polymer were completely disentangled a power law of 1 would be expected.

Thus, entanglement modification does not explain the observed decrease in viscosity and, as shown in Fig. 5a, an increase in the free volume produces the effect. This decrease manifests itself through the Rouse time scale and its subsequent relation to the monomeric friction factor in an unusual manner.

We have discussed how free volume as well as certain configurational changes to the linear polymer must account for the viscosity decrease. The exact mechanism is as yet unknown and is a challenge for theoreticians. However, it is clear that nanoparticles do not contribute hydrodynamically to the viscosity in the traditional sense predicted by Einstein; rather, they produce a change in the conformation and free volume of the polymer to decrease the viscosity. In other words, the system's properties change with the addition of nanoparticles, and different physics come into play to produce the decrease in viscosity.

METHODS

Blends of the tightly crosslinked nanoparticles with linear PS were prepared through co-dissolution in THF and rapid precipitation with methanol, followed by drying in vacuum at 40 °C for at least a week to ensure complete solvent removal. The powder was then moulded by compression under vacuum in a pellet press to ensure that no trapped air was in a sample. The samples were aged at 120–170 °C under vacuum for several hours to ensure homogeneity. The 8-mm diameter discs were placed on the 8-mm parallel plates fixture of a Rheometrics ARES rheometer set at a gap of about 0.5 mm. Various temperatures were used and the strain during the dynamic shear test was kept small enough to ensure that all response was in the linear viscoelastic region. T_gs were determined with a TA Instruments modulated differential calorimeter model Q1000. Sample sizes on the order of 4 mg were used and the instrument was fully calibrated.

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Competing financial interests

The authors declare that they have no competing financial interests.

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